15-663 Computational Photography, Fall 2026

Assignment 3 Report

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1 Bilateral filtering

I implemented piecewise bilateral filtering using Gaussian filtering at sampled guide intensities and linear interpolation between adjacent samples. Each RGB channel is processed separately. The lamp images are processed at 1536×1024.

For detail transfer, I used $\epsilon = 0.02$. The artifact mask compares linear luminance after scaling ambient values by 200/1600 to match ISO. The shadow and specularity thresholds are 0.01 and 0.95. I applied opening, closing and dilation with a 5×5 kernel, followed by Gaussian smoothing with $\sigma = 2$.

Method	σ_s (pixels)	σ_r
Bilateral	8	0.10
Joint bilateral	2	0.05
Detail transfer	8	0.10
Masked result	8	0.10

Table 1: Selected settings. Detail transfer and masking use $(8, 0.10)$ for all their intermediate bilateral filters.

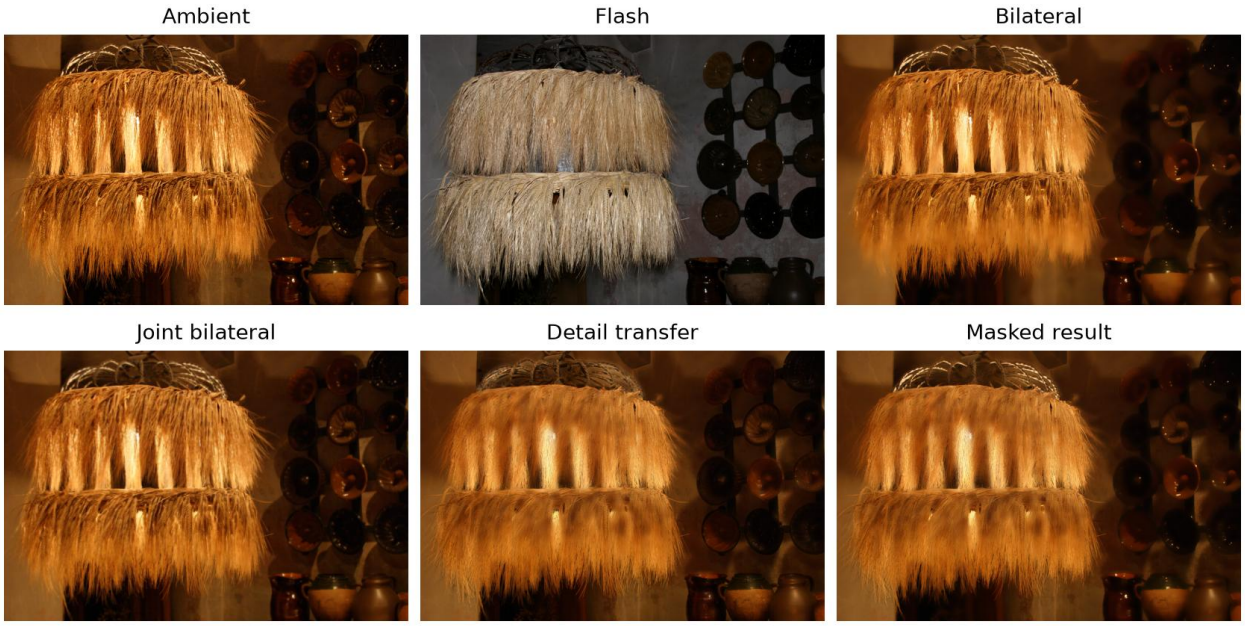


Figure 1: The four filtering results at their selected settings.

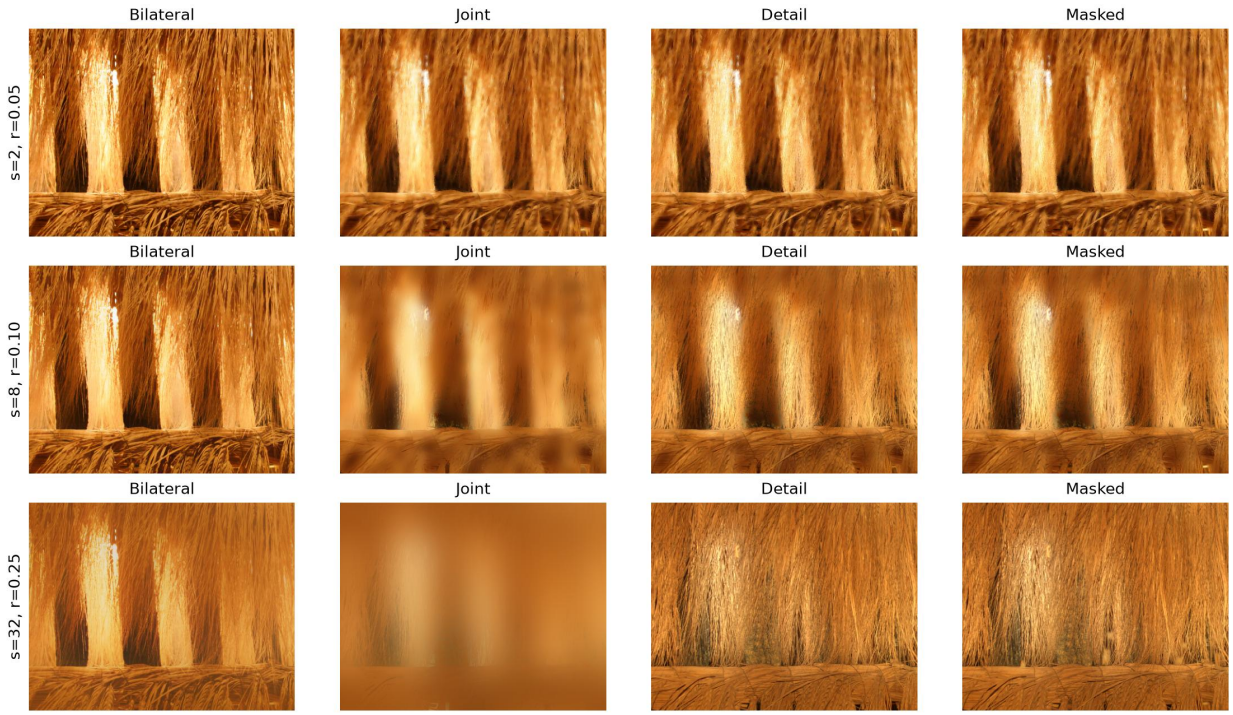


Figure 2: The same crop with three parameter settings. Rows specify (σ_s, σ_r); columns compare all four methods.

The narrow setting retains detail but leaves more noise. The widest setting smooths away the lamp's illumination structure. Basic bilateral filtering keeps strong ambient edges but cannot recover missing strands. Joint bilateral filtering uses cleaner edges, although $(8, 0.10)$ already blurs too much here; I preferred $(2, 0.05)$ when using it alone.

Detail transfer restores the flash texture while preserving the warm appearance. The middle setting balances smoothing and texture, while the widest setting changes the appearance of the bright regions too much. Masking falls back to basic bilateral filtering around flash shadows, especially above the lamp and on the background objects. It reduces transferred artifacts but also loses some detail in those regions.



Figure 3: Signed differences at the common setting $(8, 0.10)$, averaged over RGB. Red is positive and blue is negative; display limits are ± 0.08. Detail transfer mainly changes fine texture, while masking changes the detected artifact regions.

2 Gradient-domain processing

2.1 Reconstruction

I used forward differences for the gradient and backward differences for divergence. The maximum interior difference between $\operatorname{div}(\nabla I)$ and the five-point Laplacian was 4.44×10^{-16}.

With zero initialization and the original image on all four boundaries, reconstruction at tolerance 10^{-6} had maximum error 1.37×10^{-6}. The experiments below use 452×319 images and an absolute residual norm stopping criterion, separately for each channel.

Tolerance	Iteration limit	Max. iterations used	Max. error	RMSE
10^{-6}	100	100	6.34×10^{-1}	2.05×10^{-1}
10^{-3}	4000	657	1.08×10^{-3}	3.22×10^{-4}
10^{-6}	4000	1122	1.37×10^{-6}	3.41×10^{-7}

Table 2: A limit of 100 iterations stops too early. Tighter tolerance improves reconstruction.

2.2 Gradient fusion

I computed coherence using the absolute dot product, with zero coherence when a gradient has zero magnitude. Saturation weights are $w_s = [1 + \tanh(\sigma(F - \tau_s))]/2$. I then blended the ambient and flash gradients using the coherence and saturation weights and integrated their divergence.

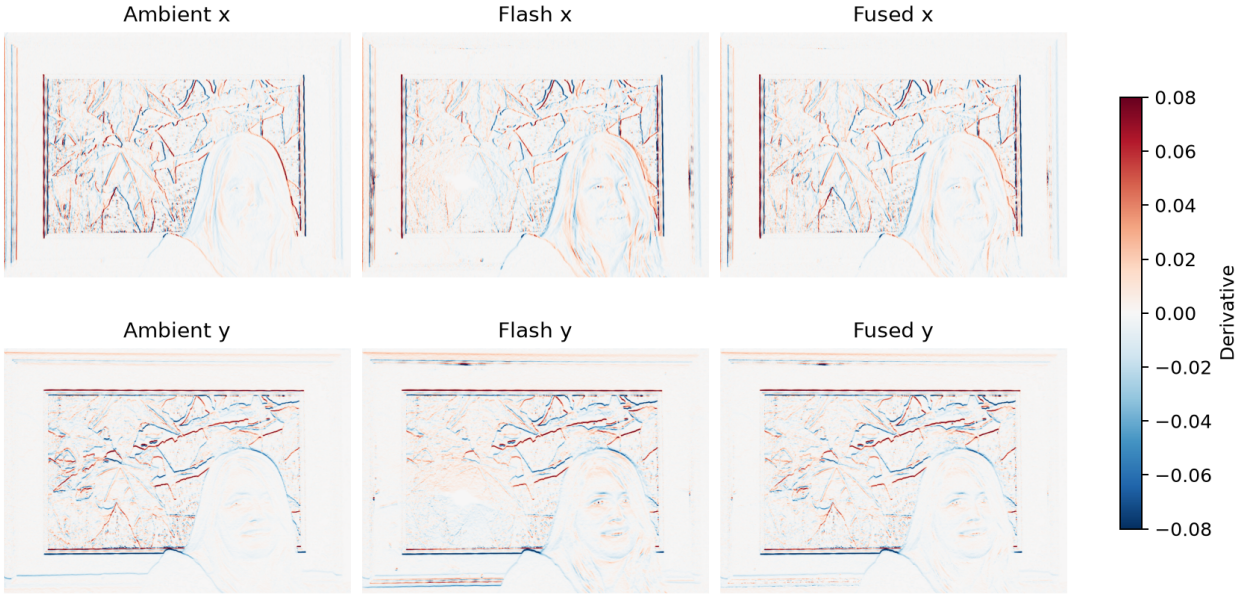


Figure 4: Ambient, flash and fused gradient components at $\sigma = 40$, $\tau_s = 0.7$. Values are averaged over RGB for display, with the same ± 0.08 scale.



Figure 5: Parameter comparison with average boundary values and average initialization.

I chose $\sigma = 40$ and $\tau_s = 0.7$. Compared with $\tau_s = 0.9$ or 0.95, the lower threshold removes more of the flash hotspot. A slope of 10 leaves a broader glow; 80 gives little improvement over 40.

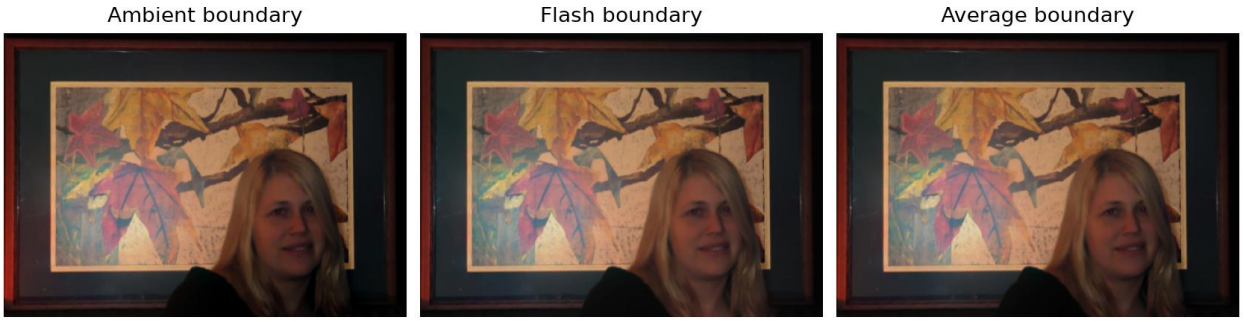


Figure 6: Changing the boundary values changes the overall brightness and color. I chose the ambient boundary to keep the darker background.

With the ambient boundary fixed, zero, ambient, flash and average initialization took at most 1083, 1076, 1076 and 1002 iterations per channel, respectively. Each result differed from zero initialization by less than 1.5×10^{-6}, so initialization changed convergence speed without visibly changing the result.

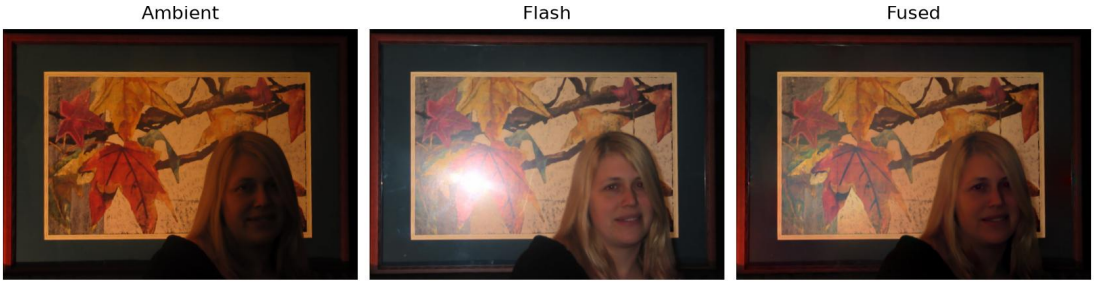


Figure 7: Final fusion at 905×638, using ambient boundary values and initialization. The face is brighter and the hotspot is reduced, though a weak color change remains on the painting.

3 Own photographs

Both pairs were captured on a tripod at f/5.6 with fixed focus and white balance. I cropped the JPEGs and aligned each pair using SIFT and a similarity transform.

Denoising. The ambient and flash images use ISO 1600 and 200, respectively, both at 1/60 s. I used $\sigma_s = 6$, $\sigma_r = 0.08$, $\epsilon = 0.02$ and the Part 1 mask thresholds. The result reduces noise on the cards while keeping the warm lighting; the rear fabric remains soft from focus.

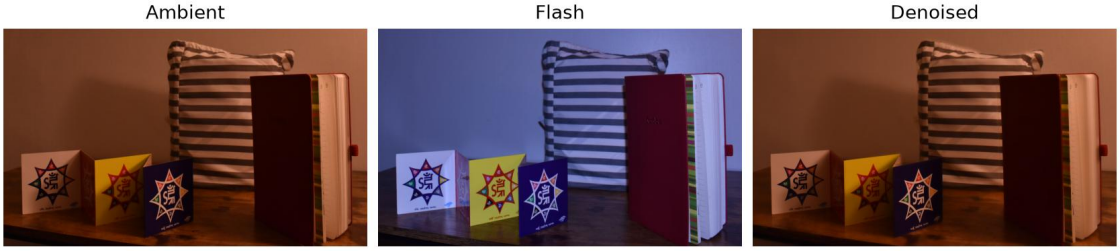


Figure 8: Desk pair and masked detail-transfer result.

Gradient fusion. Both images use ISO 200, with 1/8 s for ambient and 1/60 s for flash. I used $\sigma = 40$, $\tau_s = 0.9$, ambient boundary values and initialization, and tolerance 10^{-6}. Fusion reveals the dark cards and fabric while retaining more of the warm background. Some color shifts remain. The frosted mug produces little glare, so this pair mainly shows lighting fusion.



Figure 9: Mug pair and gradient-domain result.